Advanced CYBER SECURITY & DEFENSE PRACTITIONER

Original Professional Training Program

90 hours (4.5 months) • Beginner-Friendly • 100% Free/Open-Source Lab Stack • Theory + Hands-On Practice

TOTAL COURSE DURATION: 90 HOURS

Practical / Hands-On Labs

Course Positioning	Practical cybersecurity foundations with defensive, assessment and incident-response skills
Lab Philosophy	Use only free/open-source tools, local virtual machines, synthetic data and authorized targets
Recommended Level	Beginner to early-intermediate
Suggested Delivery	Instructor-led theory + guided labs + mini assessments + capstone

Important Course Positioning

This is an independent cybersecurity training program. It is not an official CompTIA course, is not a CompTIA certification program, and is not affiliated with or endorsed by CompTIA.

The curriculum was designed after reviewing the publicly available CompTIA Security+ SY0-701 exam-objective framework, but the module titles, sequencing, explanations, practical exercises, examples and lab activities in this document are original and intentionally restructured for hands-on learning. Students seeking the Security+ certification should consult CompTIA's current official exam objectives separately.

Recommended Course Duration: 90 Hours — 45 Hours Theory + 45 Hours Practical / Hands-On Labs.

Course Overview

This program teaches how modern security controls work, how common attacks are investigated, how systems and networks are hardened, and how a junior security practitioner can collect evidence, detect suspicious behavior and respond to incidents. The emphasis is on learning by doing rather than memorizing terminology.

Learning Outcomes

- Explain core cybersecurity principles, security objectives and common control categories.
- Build a safe personal cyber range using VirtualBox/VMware Player, Kali Linux and intentionally vulnerable systems.
- Perform authorized reconnaissance, service discovery and basic vulnerability assessment.
- Analyze network traffic with Wireshark and investigate logs with command-line and SIEM-style tools.
- Apply Linux and Windows security hardening techniques in a controlled lab.
- Understand authentication, passwords, hashing, MFA concepts, authorization and access governance.
- Apply practical cryptography concepts and create/test certificates in a local lab.
- Understand secure web application behavior and investigate common web weaknesses using OWASP tools.
- Work with firewalls, segmentation, secure protocols, VPN concepts and endpoint protections.
- Use free/open-source tools for vulnerability assessment, detection, malware triage and digital forensics.
- Build incident timelines, classify alerts, preserve evidence and document response actions.
- Understand risk, policies, privacy, business continuity, disaster recovery and security governance.
- Use Python and AI locally for defensive analysis and security workflow assistance.
- Complete an end-to-end defensive security capstone.

Free Lab Stack

Area	Free / Community Lab Tool
Virtualization	VirtualBox or VMware Workstation Player (where available)
Security OS	Kali Linux
Linux Server	Ubuntu Server
Windows Lab	Windows evaluation VM where legally available; alternatively use an existing licensed Windows system
Vulnerable Targets	OWASP Juice Shop, DVWA, Metasploitable 2/3 where legally obtained
Network Analysis	Wireshark, tcpdump

Discovery	Nmap
Web Testing	OWASP ZAP, Burp Suite Community Edition
Vulnerability Assessment	Greenbone/OpenVAS Community Edition, Nuclei
Password/Hash Learning	John the Ripper, Hashcat
Forensics	Autopsy, Volatility
Detection/Endpoint	Wazuh
Log/SIEM Practice	Wazuh and Elastic/OpenSearch components where practical; local sample logs
Threat Analysis	YARA, ClamAV, Ghidra
Automation	Python 3, Bash, PowerShell
Containers	Docker Engine/Community Edition, Trivy
Threat Intelligence	MITRE ATT&CK website/datasets, MISP Community Edition
AI	Ollama with a locally available model, plus Python-based security utilities

Note: Some tools have optional commercial editions or cloud services. The labs in this curriculum are designed so that the learning activities can be completed using free/community/local options.

Recommended Lab Topology

- Kali Linux — analyst/security testing workstation
- Ubuntu Server — server hardening, logs and services
- Windows VM — Windows security, PowerShell and event-log exercises
- OWASP Juice Shop/DVWA — intentionally vulnerable web application
- Metasploitable — isolated vulnerable target for controlled assessment
- Wazuh — security monitoring and endpoint/log analytics
- Optional Docker host — container security labs
- Host-only or isolated virtual network — no unauthorized Internet scanning

Course Roadmap

Module	Module Name	Core Focus
01	Security Mindset & Cyber Foundations	Principles, assets, threats, controls
02	Networking for Security Practitioners	TCP/IP, DNS, HTTP, segmentation
03	Build Your Cyber Range	Virtual machines, isolation, snapshots
04	Threat Discovery & Reconnaissance	OSINT, passive/active discovery
05	Network Mapping & Exposure Review	Nmap, service enumeration
06	Weakness Assessment & Risk	Vulnerability concepts, CVEs, scanning
07	Identity, Authentication & Access	Accounts, permissions, MFA, PAM
08	Applied Cryptography & PKI	Hashing, encryption, TLS, certificates
09	Secure System Administration	Linux/Windows hardening
10	Web & API Security Essentials	HTTP, sessions, common weaknesses
11	Network Defense & Secure	Firewalls, IDS/IPS, VPN, secure

	Connectivity	protocols
12	Cloud, Virtualization & Container Security	Shared responsibility, Docker, image scanning
13	Wireless, Mobile & IoT Protection	Wi-Fi security, device risks, IoT
14	Security Monitoring & Detection	Logs, Wazuh, alert analysis
15	Malware & Threat Analysis Basics	Static triage, YARA, hashes
16	Digital Forensics & Evidence Handling	Disk/memory evidence, timelines
17	Incident Response & Recovery	Triage, containment, eradication, recovery
18	Vulnerability & Penetration Testing Workflow	Authorized assessment lifecycle
19	Risk, Governance, Privacy & Resilience	Policies, risk, BCP/DR, privacy
20	Security Automation & AI Assistance	Python, local AI, analyst workflows
21	Integrated Blue-Team Challenge	Multi-stage investigation
22	Final Capstone Project	Build, secure, monitor and investigate a mini environment

Recommended Duration: 90 Hours (45 Theory + 45 Practical). The module hours include instructor-led explanation, demonstrations, guided labs, evidence review and short assessments.

Module	Module Name	Hours	Primary Emphasis
01	Security Mindset & Cyber Foundations	3	Foundations, assessment & core security
02	Networking for Security Practitioners	4	Foundations, assessment & core security
03	Build Your Cyber Range	3	Foundations, assessment & core security
04	Threat Discovery & Reconnaissance	4	Foundations, assessment & core security
05	Network Mapping & Exposure Review	4	Foundations, assessment & core security
06	Weakness Assessment & Risk	4	Foundations, assessment & core security
07	Identity, Authentication & Access	4	Foundations, assessment & core security
08	Applied Cryptography & PKI	4	Foundations, assessment & core security
09	Secure System Administration	4	Foundations, assessment & core security
10	Web & API Security Essentials	5	System, web, network & infrastructure defense
11	Network Defense & Secure Connectivity	4	System, web, network & infrastructure defense
12	Cloud, Virtualization & Container Security	4	System, web, network & infrastructure defense
13	Wireless, Mobile & IoT Protection	3	System, web, network & infrastructure defense
14	Security Monitoring & Detection	5	Monitoring, malware, forensics & response
15	Malware & Threat Analysis Basics	4	Monitoring, malware, forensics & response
16	Digital Forensics & Evidence Handling	4	Monitoring, malware, forensics & response
17	Incident Response & Recovery	5	Monitoring, malware, forensics & response
18	Vulnerability & Penetration Testing Workflow	5	Monitoring, malware, forensics & response
19	Risk, Governance, Privacy & Resilience	4	Governance, automation & AI
20	Security Automation & AI Assistance	4	Governance, automation & AI
21	Integrated Blue-Team Challenge	4	Integrated blue-team investigation

22	Final Capstone Project	5	End-to-end capstone

TOTAL: 90 HOURS

Detailed Module Content

Module 01: Security Mindset & Cyber Foundations — 3 Hours

Theory Concepts

- Security objectives: confidentiality, integrity, availability and resilience
- Assets, attack surface, exposure, threat, vulnerability, exploit and risk
- Administrative, technical and physical protection measures
- Preventive, detective, corrective, deterrent and compensating controls
- Security principles: least privilege, defense in depth, secure defaults and zero trust concepts
- Basic security terminology, roles and ethical boundaries

Hands-On Lab Objective

Create a personal asset inventory for the lab and map each asset to threats and controls. Build a simple risk register with likelihood, impact and treatment.

Practical Lab Activities

1. Define five lab assets and their owners/purpose.
2. Identify threats and vulnerabilities for each asset.
3. Create a risk matrix in a spreadsheet.
4. Map controls to the identified risks.
5. Present one risk-treatment decision.

Free / Community Tools

LibreOffice Calc, VirtualBox/VMware, Kali Linux

Expected Learning Outcome

Students can explain core security concepts and produce a basic risk register.

Module 02: Networking for Security Practitioners — 4 Hours

Theory Concepts

- IPv4/IPv6, MAC addresses, ports and protocols
- TCP vs UDP and connection behavior
- DNS, DHCP, ARP, ICMP, HTTP/HTTPS and SSH
- Subnets, routing, NAT and segmentation
- Common network attack surfaces and defensive boundaries

Hands-On Lab Objective

Observe real traffic generated by the student's own lab machines and identify protocols, endpoints and connection patterns.

Practical Lab Activities

6. Capture DNS and HTTP/HTTPS traffic in Wireshark.
7. Use tcpdump for command-line capture.
8. Inspect TCP handshakes and DNS queries.
9. Create a simple network diagram.

10. Identify which services should be restricted.

Free / Community Tools

Wireshark, tcpdump, ip/ss/netstat, Nmap

Expected Learning Outcome

Students can read basic network traffic and explain why segmentation and secure protocols matter.

Module 03: Build Your Cyber Range — 3 Hours

Theory Concepts

- Virtual machines, snapshots and isolated networks
- Host-only/NAT networking
- Safe vulnerable targets
- Lab reset and evidence preservation
- Rules of engagement and authorization

Hands-On Lab Objective

Build an isolated multi-VM cyber range without exposing vulnerable machines to the public Internet.

Practical Lab Activities

11. Install two or more Linux/Windows VMs.
12. Create an isolated virtual network.
13. Take clean snapshots.
14. Deploy an intentionally vulnerable web application.
15. Test connectivity only inside the lab.
16. Document IP addresses and recovery steps.

Free / Community Tools

VirtualBox or VMware, Kali Linux, Ubuntu, OWASP Juice Shop

Expected Learning Outcome

Students have a repeatable, isolated lab for the remainder of the course.

Module 04: Threat Discovery & Reconnaissance — 4 Hours

Theory Concepts

- Passive vs active reconnaissance
- OSINT concepts and source validation
- DNS records and domain relationships
- Technology identification
- Attack surface mapping

Hands-On Lab Objective

Perform reconnaissance against instructor-provided or self-owned lab assets only.

Practical Lab Activities

17. Collect public information about a permitted lab domain.
18. Use DNS tools to identify records.
19. Map technologies and exposed services.
20. Record findings with evidence.

21. Separate verified facts from assumptions.

Free / Community Tools

whois, dig, nslookup, Amass (passive features), theHarvester

Expected Learning Outcome

Students can build an evidence-based attack-surface map without crossing authorization boundaries.

Module 05: Network Mapping & Exposure Review — 4 Hours

Theory Concepts

- Host discovery and port scanning
- Service/version identification
- Scan timing and scope
- False positives and validation
- Reducing unnecessary exposure

Hands-On Lab Objective

Map the isolated lab network and validate the discovered services.

Practical Lab Activities

22. Run controlled Nmap host discovery.
23. Perform TCP service enumeration.
24. Compare scan results with actual services.
25. Identify unnecessary ports.
26. Write a short exposure report.

Free / Community Tools

Nmap, netcat, ss

Expected Learning Outcome

Students can perform basic authorized network discovery and explain scan results.

Module 06: Weakness Assessment & Risk — 4 Hours

Theory Concepts

- Vulnerability vs configuration weakness
- CVEs, severity concepts and exploitability
- Authenticated vs unauthenticated assessment
- False positives and manual validation
- Remediation prioritization

Hands-On Lab Objective

Assess intentionally vulnerable systems and translate scanner findings into practical remediation.

Practical Lab Activities

27. Run a community vulnerability scanner against a lab target.
28. Review sample CVE information.
29. Manually validate selected findings.
30. Rate business impact using a simple matrix.
31. Recommend remediation and retest.

Free / Community Tools

Greenbone/OpenVAS Community Edition, Nmap, Nuclei

Expected Learning Outcome

Students can distinguish scanner output from verified security findings and propose remediation.

Module 07: Identity, Authentication & Access — 4 Hours

Theory Concepts

- Identity lifecycle and account management
- Authentication factors and MFA concepts
- Authorization and permissions
- Least privilege and privilege boundaries
- Password storage, salting and password policies

Hands-On Lab Objective

Create users and permission boundaries in Linux and Windows, then investigate weak access configurations.

Practical Lab Activities

32. Create standard and privileged accounts.
33. Apply Linux file permissions and sudo rules.
34. Review Windows local-user/security settings where available.
35. Demonstrate password hashing concepts with test data.
36. Document access-control improvements.

Free / Community Tools

Linux users/groups, chmod/chown, sudo, PowerShell, John the Ripper (test hashes only)

Expected Learning Outcome

Students can configure basic access controls and explain authentication vs authorization.

Module 08: Applied Cryptography & PKI — 4 Hours

Theory Concepts

- Symmetric and asymmetric encryption
- Hash functions and integrity
- Digital signatures
- Certificates and trust chains
- TLS concepts and secure key handling

Hands-On Lab Objective

Generate test keys and certificates and inspect a local TLS connection.

Practical Lab Activities

37. Create a local CA for the lab.
38. Issue a test certificate.
39. Generate and compare hashes.
40. Create and verify a digital signature.
41. Inspect TLS certificate details.
42. Discuss key rotation and protection.

Free / Community Tools

OpenSSL, sha256sum, GPG

Expected Learning Outcome

Students can explain and demonstrate practical cryptography without relying on paid services.

Module 09: Secure System Administration — 4 Hours

Theory Concepts

- Linux hardening principles
- Windows security configuration
- Patch management
- Service minimization
- Host firewalls and secure remote administration
- Logging and time synchronization

Hands-On Lab Objective

Harden an Ubuntu server and a Windows lab endpoint, then verify the changes.

Practical Lab Activities

43. Disable unnecessary services.
44. Configure UFW on Linux.
45. Review Windows Firewall rules.
46. Apply secure SSH settings.
47. Create a basic hardening checklist.
48. Generate before/after evidence.

Free / Community Tools

Ubuntu, UFW, OpenSSH, Windows Firewall, PowerShell

Expected Learning Outcome

Students can apply baseline hardening and document security improvements.

Module 10: Web & API Security Essentials — 5 Hours

Theory Concepts

- HTTP request/response model
- Cookies, sessions and authentication
- Input validation and output encoding
- Access-control weaknesses
- Injection and client-side attack concepts
- API security basics

Hands-On Lab Objective

Investigate a deliberately vulnerable web application and identify security weaknesses without attacking real systems.

Practical Lab Activities

49. Intercept a lab web request.
50. Explore authentication/session behavior.
51. Test harmless inputs in the vulnerable app.

52. Identify access-control weaknesses.
53. Document evidence and remediation.
54. Repeat after a fix.

Free / Community Tools

OWASP ZAP, Burp Suite Community, OWASP Juice Shop, DVWA

Expected Learning Outcome

Students can understand and test common web security weaknesses in a controlled environment.

Module 11: Network Defense & Secure Connectivity — 4 Hours

Theory Concepts

- Firewall concepts and rule design
- IDS/IPS concepts
- Network segmentation and DMZs
- Secure vs insecure protocols
- VPN fundamentals
- Email and DNS security concepts

Hands-On Lab Objective

Design and implement a small defensive network policy for the lab.

Practical Lab Activities

55. Create firewall allow/deny rules.
56. Test permitted and blocked traffic.
57. Capture evidence of blocked connections.
58. Compare HTTP vs HTTPS and Telnet vs SSH.
59. Build a simple segmented-network design.

Free / Community Tools

UFW, iptables/nftables concepts, Wireshark, OpenVPN/WireGuard for lab study

Expected Learning Outcome

Students can design basic network controls and validate them with traffic evidence.

Module 12: Cloud, Virtualization & Container Security — 4 Hours

Theory Concepts

- Cloud shared-responsibility concept
- Identity and access in cloud environments
- Virtual machine security
- Container images and registries
- Secrets management
- Container isolation and runtime risks

Hands-On Lab Objective

Secure a local Docker workload and scan its image for known weaknesses.

Practical Lab Activities

60. Build a minimal container.

61. Run it with reduced privileges.
62. Inspect image layers.
63. Scan the image for vulnerabilities.
64. Remove unnecessary packages.
65. Document secure container practices.

Free / Community Tools

Docker Community Edition, Trivy, Docker Scout alternatives where free/local use is available

Expected Learning Outcome

Students can identify common cloud/container risks and apply basic container hardening.

Module 13: Wireless, Mobile & IoT Protection — 3 Hours

Theory Concepts

- Wi-Fi security generations and encryption
- Authentication and rogue access-point risks
- Mobile device attack surface
- IoT constraints and default credentials
- Device segmentation and monitoring

Hands-On Lab Objective

Analyze a captured wireless dataset or instructor-provided lab material without disrupting real networks.

Practical Lab Activities

66. Inspect a permitted capture file.
67. Identify SSID/security characteristics.
68. Explain WPA2/WPA3 security concepts.
69. Create an IoT segmentation diagram.
70. Build a device security checklist.

Free / Community Tools

Wireshark, Aircrack-ng suite with lab captures, Kismet where supported

Expected Learning Outcome

Students can assess wireless/IoT security concepts safely without attacking public networks.

Module 14: Security Monitoring & Detection — 5 Hours

Theory Concepts

- Why logs matter
- Event sources and normalization
- Authentication and process telemetry
- Detection logic and alert quality
- False positives and triage

Hands-On Lab Objective

Deploy or use Wazuh in a local lab and investigate generated security events.

Practical Lab Activities

71. Connect a Linux endpoint to Wazuh.

72. Generate controlled authentication events.
73. Review alerts and event context.
74. Create a simple detection rule or query.
75. Classify alerts and document tuning.

Free / Community Tools

Wazuh, Linux auth logs, Windows Event Logs, Sigma rule concepts

Expected Learning Outcome

Students can move from raw telemetry to a basic security alert and investigation.

Module 15: Malware & Threat Analysis Basics — 4 Hours

Theory Concepts

- Malware categories and behavior
- File hashes and reputation concepts
- Static vs dynamic analysis
- Indicators of compromise
- YARA rule fundamentals
- Safe malware-analysis boundaries

Hands-On Lab Objective

Perform static analysis of instructor-provided benign samples or safe test files.

Practical Lab Activities

76. Calculate hashes.
77. Inspect file metadata and strings.
78. Create a basic YARA rule.
79. Scan a controlled directory.
80. Document indicators.
81. Explain why unknown malware must remain isolated.

Free / Community Tools

YARA, ClamAV, strings, file, Ghidra for safe binaries

Expected Learning Outcome

Students can perform basic malware triage without executing unsafe samples.

Module 16: Digital Forensics & Evidence Handling — 4 Hours

Theory Concepts

- Evidence sources and volatility
- Acquisition vs analysis
- Chain of custody
- File-system artifacts
- Memory-analysis concepts
- Timeline reconstruction

Hands-On Lab Objective

Analyze a provided forensic image or memory sample and create an evidence timeline.

Practical Lab Activities

82. Record evidence metadata.
83. Identify relevant artifacts.
84. Extract timestamps.
85. Build a timeline.
86. Use a forensic tool to examine files.
87. Write a short findings report.

Free / Community Tools

Autopsy, Volatility 3, Plaso/log2timeline where practical

Expected Learning Outcome

Students can conduct a beginner-level forensic investigation and preserve evidence context.

Module 17: Incident Response & Recovery — 5 Hours

Theory Concepts

- Preparation, detection, analysis, containment, eradication and recovery
- Incident severity and escalation
- Evidence preservation
- Communication and documentation
- Lessons learned
- Backup and recovery concepts

Hands-On Lab Objective

Respond to a simulated compromise using logs and evidence from the earlier modules.

Practical Lab Activities

88. Receive a simulated alert.
89. Build an incident timeline.
90. Identify affected host/account.
91. Contain the lab endpoint.
92. Recommend eradication and recovery.
93. Write an incident report and lessons learned.

Free / Community Tools

Wazuh, Wireshark, Autopsy, LibreOffice

Expected Learning Outcome

Students can follow a repeatable incident-response workflow from alert to closure.

Module 18: Vulnerability & Penetration Testing Workflow — 5 Hours

Theory Concepts

- Assessment scope and authorization
- Reconnaissance, enumeration and validation
- Finding evidence
- Safe proof-of-concept principles
- Remediation verification
- Professional reporting

Hands-On Lab Objective

Perform a complete authorized security assessment of the intentionally vulnerable lab.

Practical Lab Activities

94. Define scope and rules of engagement.
95. Discover assets.
96. Enumerate services.
97. Assess web/application weaknesses.
98. Validate selected findings safely.
99. Produce an executive summary and technical report.
100. Retest remediation.

Free / Community Tools

Nmap, Nuclei, OWASP ZAP, Greenbone/OpenVAS, Kali Linux

Expected Learning Outcome

Students can execute and report a controlled assessment without confusing scanning with exploitation.

Module 19: Risk, Governance, Privacy & Resilience — 4 Hours

Theory Concepts

- Security policies and standards
- Risk identification and treatment
- Data classification
- Privacy principles
- Vendor and third-party risk
- Business continuity and disaster recovery
- Backup strategies and recovery testing

Hands-On Lab Objective

Create a mini security program for a fictional organization.

Practical Lab Activities

101. Write an acceptable-use policy outline.
102. Create a data classification table.
103. Build a risk register.
104. Create a basic BCP/DR scenario.
105. Define backup/recovery objectives.
106. Map selected controls to a recognized framework.

Free / Community Tools

LibreOffice Writer/Calc, NIST CSF reference material, CIS Controls reference material

Expected Learning Outcome

Students can connect technical security work with organizational risk and governance.

Module 20: Security Automation & AI Assistance — 4 Hours

Theory Concepts

- Python for security automation
- Parsing logs and structured data

- API concepts
- Detection enrichment
- Local AI and prompt safety
- Human-in-the-loop security decisions

Hands-On Lab Objective

Build a small defensive automation that reads a local log file, extracts indicators and produces a structured analyst summary.

Practical Lab Activities

107. Write a Python log parser.
108. Extract IPs/domains/hashes.
109. Apply simple validation.
110. Generate a CSV/JSON summary.
111. Optionally use Ollama for local summarization.
112. Require human verification before any response action.

Free / Community Tools

Python 3, Ollama, Pandas (optional), Regex

Expected Learning Outcome

Students can automate repetitive analyst tasks while keeping security decisions under human control.

Module 21: Integrated Blue-Team Challenge — 4 Hours

Theory Concepts

- Multi-source correlation
- Alert triage
- Threat hunting hypotheses
- Evidence-driven decisions
- Detection tuning

Hands-On Lab Objective

Investigate a simulated multi-stage attack across endpoint, network and application logs.

Practical Lab Activities

113. Start with an ambiguous alert.
114. Correlate Wazuh and network evidence.
115. Identify the probable attack path.
116. Map behaviors to MITRE ATT&CK techniques.
117. Tune a noisy detection.
118. Prepare a management summary.

Free / Community Tools

Wazuh, Wireshark, MITRE ATT&CK, Sigma concepts, Python

Expected Learning Outcome

Students demonstrate investigation, correlation, detection tuning and communication skills.

Module 22: Final Capstone Project — 5 Hours

Theory Concepts

- Security architecture basics
- Asset discovery and risk analysis
- Hardening
- Detection
- Incident response
- Reporting and presentation

Hands-On Lab Objective

Build a small organization lab and demonstrate prevention, detection and response end-to-end.

Practical Lab Activities

119. Create a 3–5 VM environment or equivalent container lab.
120. Document architecture and assets.
121. Perform an authorized exposure assessment.
122. Harden critical systems.
123. Deploy monitoring.
124. Generate controlled security events.
125. Investigate one incident.
126. Produce remediation plan.
127. Present findings and lessons learned.

Free / Community Tools

VirtualBox/VMware, Kali, Ubuntu, Wazuh, Wireshark, Nmap, OWASP Juice Shop, Python, LibreOffice

Expected Learning Outcome

A portfolio-ready cybersecurity project demonstrating practical blue-team and assessment skills.

Delivery Recommendation: 30 sessions × 3 hours, or 22–23 sessions of approximately 4 hours. Keep the capstone and blue-team challenge as dedicated practical sessions.

Practical Assessment Plan

Component	Weight	Evidence
Knowledge Checks	15%	Short quizzes after major sections
Guided Labs	25%	Completion and evidence from hands-on exercises
Investigation Challenge	20%	Analyze logs, traffic and alerts
Security Assessment Report	15%	Authorized vulnerability assessment and remediation
Final Capstone	25%	Build, secure, monitor and investigate a mini environment

Suggested Capstone Deliverables

- Lab architecture/network diagram
- Asset inventory and risk register
- Baseline hardening checklist
- Network exposure scan report
- One web-security assessment report
- Detection rule/query with test evidence
- Incident timeline and response report
- Remediation plan
- Final presentation/demo

Lab Safety & Ethics

- All scanning, exploitation and testing must be performed only against systems the learner owns or has explicit permission to test.
- Keep intentionally vulnerable systems on an isolated lab network whenever possible.
- Never use public IPs, public websites, third-party accounts or production systems for attack demonstrations.
- Use synthetic credentials, test hashes and instructor-provided samples.
- Do not execute unknown malware on a normal workstation. Use a dedicated isolated analysis environment.
- Document scope and authorization before every assessment lab.
- The purpose of offensive-security exercises is controlled learning and defensive improvement.

Recommended Course Positioning

Suggested short tagline: “LEARN → PRACTICE → DETECT → RESPOND”

The program can be marketed as an independent practical cybersecurity foundation course rather than as a replacement for any vendor certification. If the learner intends to sit for CompTIA Security+, they should separately compare their knowledge against the latest official SY0-701 objectives.

Reference Note

CompTIA's official SY0-701 exam-objectives document was used only as a high-level reference for topic coverage. This training document deliberately uses original wording, a different module sequence, additional practical emphasis, and a free/open-source lab orientation.

Official reference: CompTIA Security+ SY0-701 Certification Exam Objectives, Version 6.0 (CompTIA, 2023).